

# 5513assignment 1

student id:35742305

Tiansi Zhu

## Data Cleaning

To ensure data quality, the following cleaning steps were performed: 1. Removed rows with missing values in the `purchase_amount` variable, as these observations could not be used for analysis. 2. Verified the data types of all variables to ensure consistency.

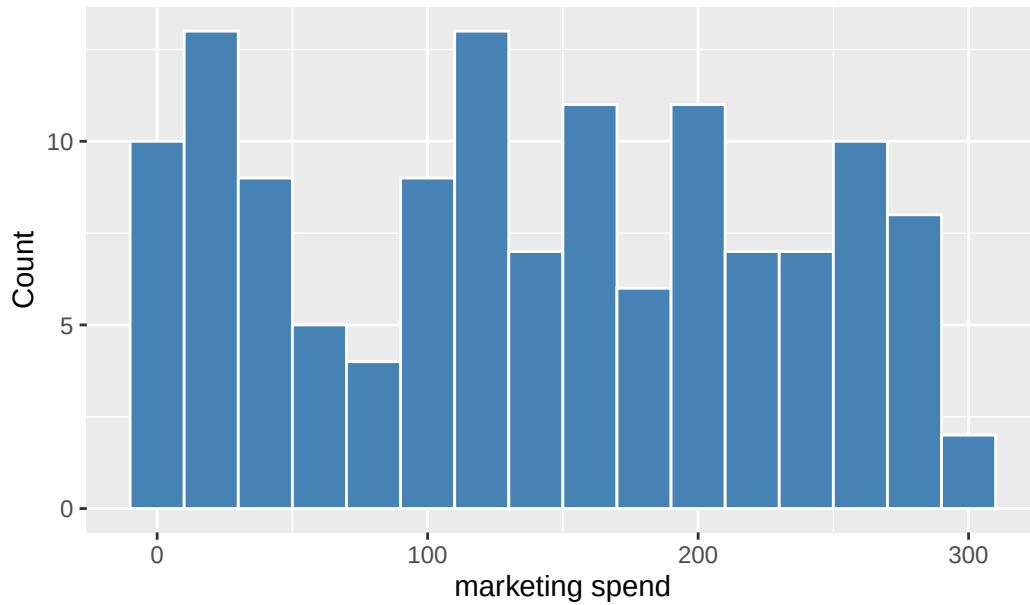
```
[1] 132    7
```

```
[1] "customer_id"      "signup_date"      "marketing_spend"  "visits"
[5] "purchase_amount"  "segment"          "notes"
```

```
spc_tbl_ [132 x 7] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ customer_id      : chr [1:132] "35742305_c0001" "35742305_c0002" "35742305_c0003" "35742305_
 $ signup_date      : chr [1:132] "13/08/2025" "2025-10-28" "2025-03-25" "2025-
03-21" ...
 $ marketing_spend: num [1:132] 128.27 126.09 4.28 270.18 156.86 ...
 $ visits          : num [1:132] 5 9 1 10 7 0 4 1 3 2 ...
 $ purchase_amount: num [1:132] 157.6 86.5 0 224.4 131.6 ...
 $ segment         : chr [1:132] "SMB" "Enterprise" "Consumer" "Enterprise" ...
 $ notes           : chr [1:132] "contacted" NA NA NA ...
- attr(*, "spec")=
.. cols(
..   customer_id = col_character(),
..   signup_date = col_character(),
..   marketing_spend = col_double(),
..   visits = col_double(),
..   purchase_amount = col_double(),
..   segment = col_character(),
..   notes = col_character()
.. )
- attr(*, "problems")=<externalptr>
```

The data contains 132 rows and 7 columns. And it contains customer\_id, signup\_date, marketing\_spend, visits, purchase\_amount, segment, notes 7 variables. It contains 132 observations each variable.

Distribution of marketing spend



Distribution of purchase amount

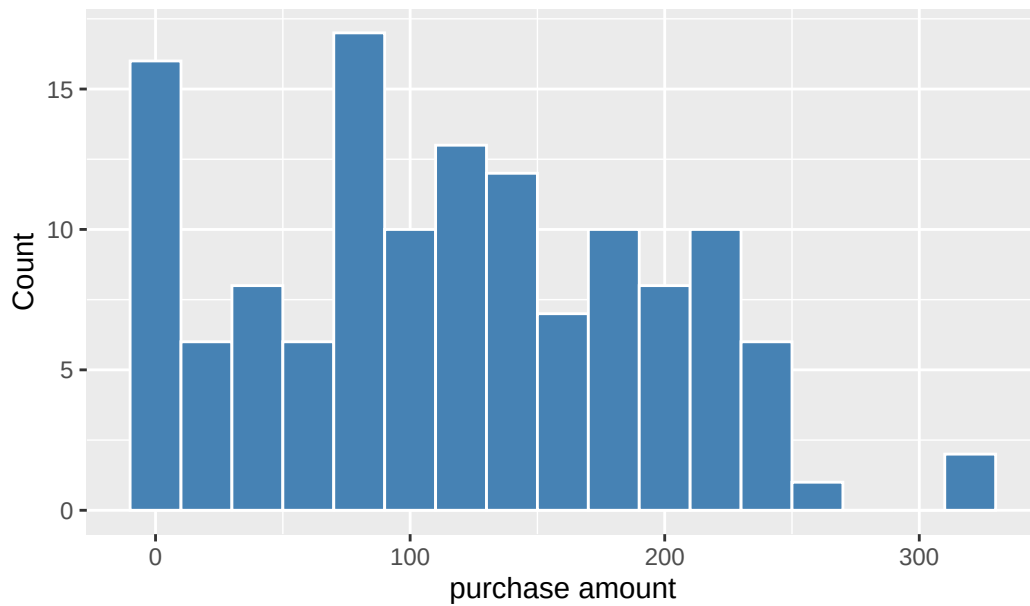


Table 1: Summary of the data.

| me-     |       |        |      |        |        |     |     |     |       |     |       |       |        |          |
|---------|-------|--------|------|--------|--------|-----|-----|-----|-------|-----|-------|-------|--------|----------|
| mean    | dian  | max    | min  | nsch   | mar-   | me- |     |     |       |     |       | me-   |        |          |
| ket-    | ket-  | ket-   | ket- | ket-   | mean   | dis | vis | ax  | mis   | vis | mean  | dian  | max    | min      |
| ing     | ing   | ing    | ing  | ing    | its    | its | its | its | its   | its | chase | chase | chase  | chase    |
| 139.424 | 43.61 | 295.57 | 0.29 | 89.057 | 355303 | 5   | 18  | 0   | 2.882 | 732 | 732   | 3195  | 12.415 | 313.34   |
|         |       |        |      |        |        |     |     |     |       |     |       |       | 0      | 76.47368 |

Table 1 shows

Call:

```
lm(formula = purchase_amount ~ marketing_spend, data = data)
```

Residuals:

| Min      | 1Q      | Median | 3Q     | Max     |
|----------|---------|--------|--------|---------|
| -114.816 | -32.624 | -7.954 | 32.163 | 154.167 |

Coefficients:

|                 | Estimate | Std. Error | t value | Pr(> t )    |
|-----------------|----------|------------|---------|-------------|
| (Intercept)     | 23.76634 | 7.89505    | 3.01    | 0.00314 **  |
| marketing_spend | 0.66382  | 0.04777    | 13.89   | < 2e-16 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.7 on 130 degrees of freedom

Multiple R-squared: 0.5976, Adjusted R-squared: 0.5945

F-statistic: 193.1 on 1 and 130 DF, p-value: < 2.2e-16

Figure 1 shows

Figure 2 shows

The Relationship Between marketing spend per customer and

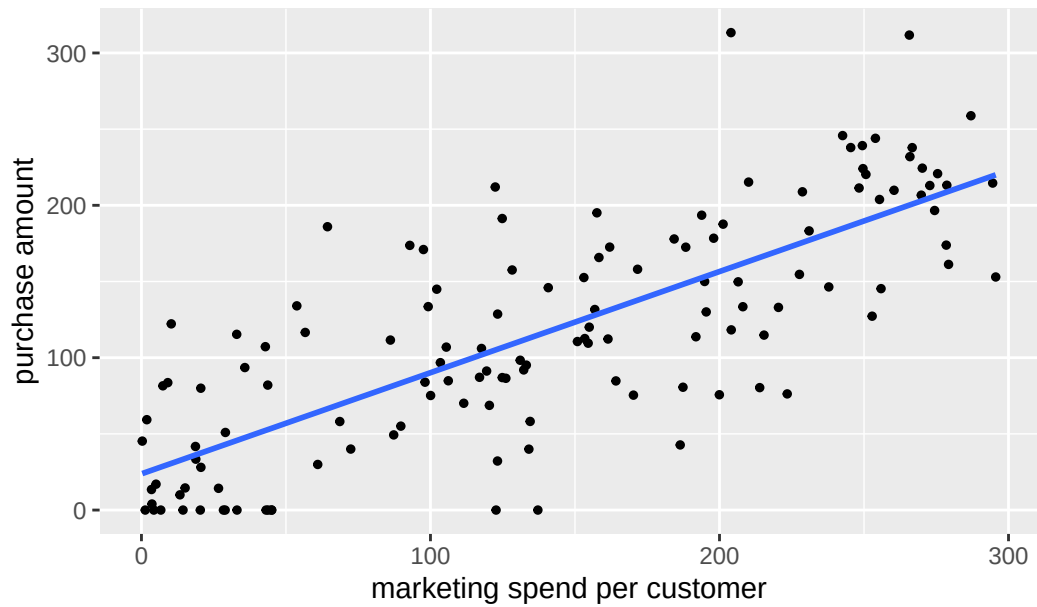


Figure 1: marketing spend per customer VS purchase amount

The Relationship Between marketing spend per customer and

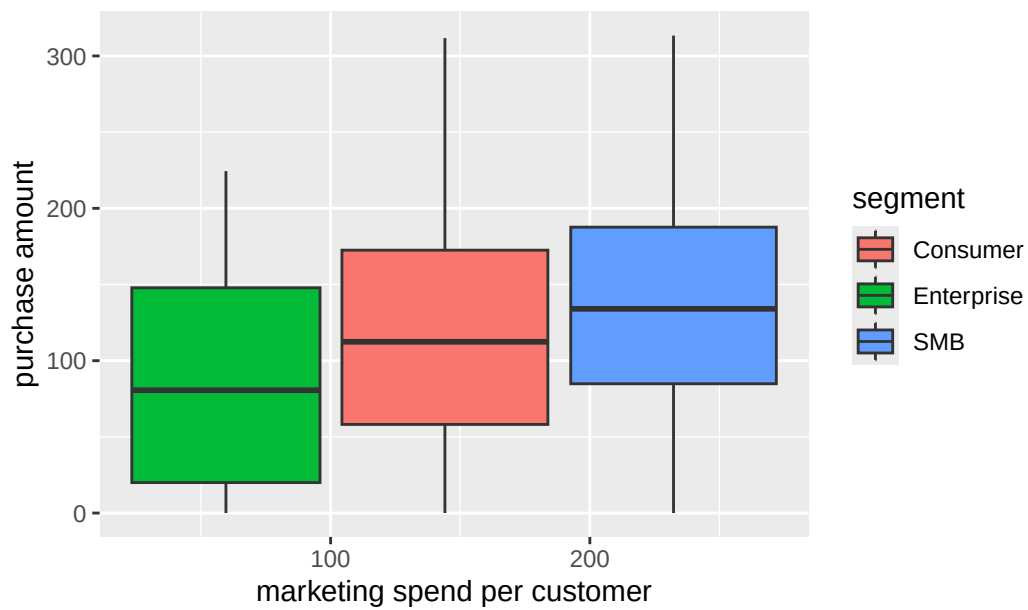


Figure 2: marketing spend per customer VS purchase amount